qp: the backtest

Above: QP.md §3 (the backtest).

Talks to:

- the strategy engine, whose `evaluate()` it calls day by day;
- the bar feeds, for bars as of a day;
- the shared record, for registers, market states and CANSLIM facts as of a day;
- the research worker, the only writer of those facts;
- qp's own files, where runs and trades live.

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1. What it is

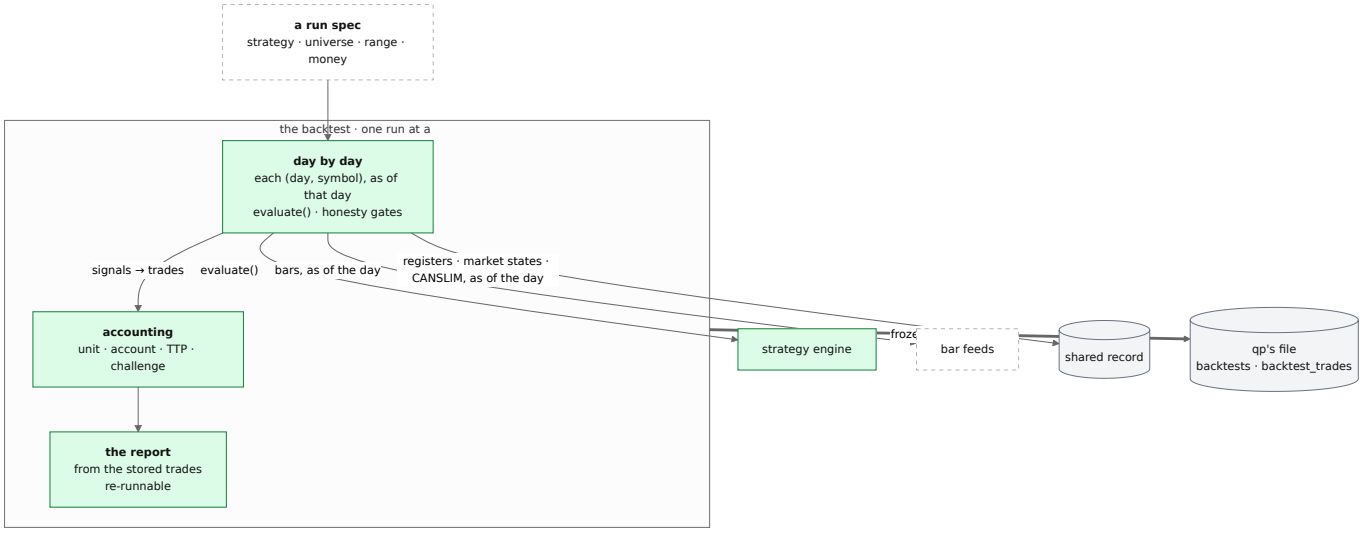
A strategy, replayed over many days on the stocks the tools actually found, on the days they found them, honestly. For every trading day in the range and every symbol in that day's universe, it calls **the same `strategy.evaluate()` the chart preview uses** (STRATEGIES.md), as of that day. There is no separate backtest evaluator, so the preview and the backtest cannot disagree.

What it owns: the run, its trades, and the report. Every run stores a frozen copy of the strategies it ran, so editing a strategy later never changes what an old result means.

What it never does: compute an indicator or a fact, or look ahead.

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2. The diagram



One run at a time (the box is small). A second start while one runs is refused with the running one's number. A run can be stopped (at its next symbol); closing the page does not stop it; if qp restarts, the run is marked interrupted. It works one (day, symbol) pair at a time, reporting the share of pairs done.

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3. Its parts

3.1 The universe

`universe.kind`	Each day evaluates
`symbols`	the same fixed list every day
`register`	one register (R1 or Shortlist) **as that tool froze it that day**
`tools`	the registers of every tool assigned to the strategy, unioned; with two books (a long and a short strategy), their tools are unioned too

`extra_symbols` adds names to a register universe; their trades are marked `_extra`, so "not on the register" stays visible.

3.2 Honesty rules (all on by default)

Rule	What it prevents
As of the day. Each (day, symbol) is evaluated with `asof` = that day; only trades **entered that day** count. Warm-up days are context, never trades	look-ahead; double-counting warm-up signals
The watchlist gate (`scan_gate`): a trade taken **before the tool had found the stock** is dropped (`scan_gate: false` shows the old numbers on purpose; no effect on a `symbols` universe)	trades no alert could have fired
The desk fill (`fill: desk`): next bar's open, levels from the decision close	the optimistic `close` fill
Costs: `cost_bps` per side (a round trip pays twice), and per-share fees with a per-order minimum	"a backtest without costs lies"
Open at the end: a position still open when its window ends is recorded as `open` with its unrealised return, never mixed into the win rate	flattering win rates
Coverage, stated: pairs evaluated, pairs whose feed returned no bars, pairs per tool, how many each filter removed	a thin feed or one dominant tool passing for a strategy that rarely fires

3.3 The settings of a run

What to test: `strategy_id` / `strategy` / `strategies` (one, or two books); `universe` (default register R1); `start` , `end` ; `tf` (default 5m); `feed` ; `view` (all / regular); `fill` (default **desk**); `days` warm-up (default 3); `rules` (session rules).

Filters and caps (opt-in):

Spec	Does
`min_rvol`	the In-Play filter: evaluate a pair only if its relative volume, measured by qp at the strategy's window start, is above this; below and unverifiable are counted apart
`canslim`	a gate on the CANSLIM letters **as known on that day**, read from the shared record; a rule naming a letter that cannot be rebuilt for a past day is refused before the run; blocked and unverifiable pairs are counted. **In the design it has a control on the page** (Q-N4)
`check_shortable`	drops shorts the broker would not lend. Alpaca keeps no borrow history, so today's flag is applied to a past day and the report says so; a name it cannot answer for is kept and named (FEEDS.md)
`rank_per_day` + metric	keep the best N signals a day. Metrics: `vwap_extension`, `tight_stop`, `reg_score`, `rvol`. **No default: an unnamed or unknown metric is an error**, because the wrong one silently reshapes the strategy
desk caps	one entry per stock per day, and a whole-run trades-per-day cap, applied in the order signals fired

Money: choose one accounting.

Block	Answers
unit	per-trade % returns
account	real money: shares from the stop so every stop-out loses the same %; equity compounds in trade order; trades with no usable stop are excluded and counted, never sized at 1 share; trades the buying power could not take are skipped and counted
TTP (prop firm)	dollar P&L for a fixed size; the min-profit rule (a winner under `min_profit_ps` $/share does not count towards the target; losses always count), tested on the position's average across its fills; every fill pays its own order fee
challenge (on an account run)	did the account reach `target_pct` **before** it ever touched `max_dd_pct`? Also walks a worst case in which every open position stops out at once

3.4 What a run reports

The report is computed afterwards from the stored trades, so it re-runs on any past backtest, and it says whether every number is in **dollars** (an account run) or **unit %**.

Section	Numbers
headline	trades, wins, losses, breakeven (exactly 0 is not a win), win rate, gross profit/loss, net profit, **profit factor** (none with no losing trade, never ∞), expectancy, average win/loss, payoff ratio, largest win/loss
R-multiples	total, average, best, worst, and the distribution in buckets; **R reads from net**, the same money as net profit; the gross total beside it
holding time	average, longest, shortest, winners vs losers, total time in positions
splits	by side, exit reason, symbol, day
equity and risk	the curve marked at exit, maximum drawdown in $ and %, its length, streaks
ratios	Sharpe, Sortino, Calmar from daily returns over every session in the range (flat days included), annualised and flagged as extrapolation on short samples
exposure	share of regular-hours minutes in a position
sample adequacy	each warning as a fact and its reason
the borrow check	states loudly when "check shortable" was ticked but nothing was answered
prop-firm detail	counted vs wasted profit under the min-profit rule

On the page, each trade opens that symbol and day on the chart with the run's own timeframe, feed, fill model and strategy; a filter narrows the trades and the filtered CSV. `rank_sweep` re-runs the per-day selection with every metric to measure which suits a strategy.

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4. What changes from today

Change	Why
Registers, market states and CANSLIM facts are read **as of a day from the shared record**	today they are read over HTTP from each tool's port, and a stopped tool needed the archive program
The **CANSLIM gate has a control** on the page	it is API-only today (Q-N4)
Past backtests are **re-run against the shared record and compared** before the switch	to prove the shared record returns the same universes and facts the HTTP path did (step D)

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5. Contracts

The code: `quant-platform/chart/backtest.py`, `report.py`, `rank_sweep.py`, `store.py`.

It owns (qp's file, `~/.qp/platform.db`)	Columns
`backtests`	id, name, spec (the run spec JSON, **including a frozen copy of the strategies it ran**), status, progress, summary, error, timestamps
`backtest_trades`	id, bt_id, date, symbol, side, entry_ts, exit_ts (empty = open at the day's end), entry, exit, ret, reason (SL, TP, trail, exit, eod, open), ctx (the frozen register card)

It reads	From
bars, as of a day	the bar feeds (FEEDS.md)
a register as that tool froze it that day	the shared record (built, off: `quantRegistersMode`)
market states and CANSLIM letters, as of a day	the shared record (research worker's facts)

Routes, behind `/quant/`	Do
`POST /api/backtest`	start; refused while another runs
`GET /api/backtest/{id}` (`trades`)	status, progress, summary, trades
`GET /api/backtests` (`brief`)	every run, newest first
`POST /api/backtest/{id}/stop`; `DELETE …`	stop at the next symbol; delete
`GET /api/backtest/{id}/report`, `…/stats.json`, `…/csv`, `…/journal.csv`	the HTML report; every number; every trade with its card; one row per trade

6. When it fails

Event	What happens
the shared record is down	a run that needs registers or facts says so and does not run; charts still work
a feed returns no bars for a pair	that pair is counted in coverage as empty; the run goes on
a CANSLIM letter cannot be rebuilt for a past day	the run is refused before it starts, naming the letter
the borrow check is asked for and Alpaca answers nothing	the trade stands as "unchecked" and the report says so loudly
qp restarts mid-run	the run is marked interrupted ("run it again")

7. Built on

The old desk: legacy/qp/BACKTEST.md.

What failed there:

ID	Fault	Prevented by
F-26	the CANSLIM gate read the entry day's own close	letters as known **before** the decision bar
F-27	an unreadable market model blocked pairs, and the failure was cached	the market state read as of a day; no cached failure
F-28	as-of earnings dropped a quarter once re-reported a year later	earnings keyed by the quarter, not the filing
F-20	`monday_high/low` and pre-market `day_open` saw the future	fixed in the library (2026-10)
F-23	Yahoo daily reach was capped at a year, so CANSLIM N/S failed on yahoo	the real reach used
F-25	`rank_sweep --backtest` found no stop on stored trades	the stop stored on every trade
Q-N4	the CANSLIM gate has no control on the page	a control on the page
—	registers read over HTTP from each tool's port; a stopped tool needed the archive	read from the shared record

8. Status

Part	Status
the engine, the universes, the honesty rules, the accounting, the report	**built** (legacy, kept)
registers from the shared record	**built, off** (`quantRegistersMode`); today qp reads each tool's port
market states and CANSLIM letters as of a day	**to build** (step D)
the CANSLIM control on the page	**to build** (Q-N4)
a re-run of past backtests against the shared record, compared	**to do**, before step D